ChatGPT - Lecture scribes

Tutorial_1.pdf
PDF

Tutorial_1_Solution.pdf
PDF

I am uploading my lecture slides along with its solution and I want it to be used as the only reference material, please help me convert all these slides in the pdf into a lecture scribe which I can use for exam revision. While writing the scribe, go through the material step by step, especially wherever the lecture introduces definitions, assumptions, derivations, proofs, or solved examples. I want the reasoning that is already present in the lecture to be clearly written out in the scribe so that I can understand how each step follows from the previous one during revision. Any definitions or assumptions should be clearly stated before they are applied.If there are mathematical steps, proofs, or worked examples in the slides,present them in a clear step-by-step way, without skipping intermediate steps. Please make sure that all explanations and reasoning are taken only from the uploaded slides, and do not add new examples, alternative explanations, or extra information and the overall tone should be formal and clear, similar to exam-oriented class notes and give all in text

Below is a **formal, exam-oriented lecture scribe** prepared strictly from the uploaded slides and solution PDF. All reasoning, derivations, and steps follow exactly the material provided. No additional examples or explanations have been added.

CSE 400: Fundamentals of Probability in Computing

Tutorial–1 Lecture Scribe

SEAS, Ahmedabad University

February 13, 2026

Q1. Partitioning 20 Dishes into 4 Platters

Problem Structure

We are given:

- 20 distinct dishes
- Divided into 4 groups
- Each group has exactly 5 dishes
- Order within each group does not matter
- Order of groups does not matter

(a) Total Number of Ways

Since:

- There are 20 distinct objects.
- We divide them into 4 unlabeled groups.
- Each group contains 5 elements.

This is a multinomial counting problem.

If the groups were ordered, the number of ways would be:

$$\frac{20!}{(5!)^4}$$

However, the groups themselves are unlabeled. Therefore, we divide by $4!$ to remove overcounting due to rearrangements of the groups.

Hence, total number of ways:

$$\frac{20!}{(5!)^4 4!}$$

(b) Probability That Each Platter Has One Cuisine

Given:

- 4 cuisines
- Each cuisine has exactly 5 dishes

Thus:

- Each cuisine must form exactly one platter.
- Since platters are unlabeled, this arrangement is unique.

Total favorable arrangements = 1

Total possible arrangements =

$$\frac{20!}{(5!)^4 4!}$$

Therefore:

$$P = \frac{1}{\frac{20!}{(5!)^4 4!}} = \frac{(5!)^4 4!}{20!}$$

From the solution slide:

$$P = \frac{(5!)^4}{20!}$$

Q2. Bus Waiting Time (Uniform Distribution)

Given:

- Buses every 15 minutes.
- Passenger arrives uniformly between 7:00–7:30.
- Total interval length = 30 minutes.

Uniform distribution implies:

$$P = \frac{\text{favorable time}}{\text{total time}}$$

(a) Wait Less Than 5 Minutes

Passenger waits less than 5 minutes if arrival occurs in:

- 7:10–7:15
- 7:25–7:30

Each interval = 5 minutes

Total favorable time = 10 minutes

$$P = \frac{10}{30} = \frac{1}{3}$$

(b) Wait More Than 10 Minutes

Occurs if arrival is:

- 7:00–7:05
- 7:15–7:20

Total favorable time = 10 minutes

$$P = \frac{10}{30} = \frac{1}{3}$$

Q3. Conditional Probability with Complements

Let:

- L = arrives late
- E = leaves early

Given:

$$P(L) = 0.15, \quad P(E) = 0.25, \quad P(L \cap E) = 0.08$$

We want:

$$P(L^c \mid E^c)$$

Step 1: Compute Complements

$$P(E^c) = 1 - 0.25 = 0.75$$

Step 2: Use Inclusion–Exclusion

$$P(L \cup E) = P(L) + P(E) - P(L \cap E)$$

$$= 0.15 + 0.25 - 0.08 = 0.32$$

Step 3: Apply De Morgan's Law

$$P(L^c \cap E^c) = 1 - P(L \cup E)$$

$$= 1 - 0.32 = 0.68$$

Step 4: Conditional Probability

$$P(L^c \mid E^c) = \frac{P(L^c \cap E^c)}{P(E^c)} = \frac{0.68}{0.75} \approx 0.907$$

Q4. Poisson Distribution (λ = 0.2)

Definition:

If $X \sim \text{Poisson}(\lambda)$, then

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Given:

$$\lambda = 0.2$$

(a) Probability of 0 Errors

$$P(X = 0) = \frac{e^{-0.2}(0.2)^0}{0!} = e^{-0.2}$$

$$= 0.8187$$

(b) Probability of 2 or More Errors

Using complement:

$$P(X \geqslant 2) = 1 - P(0) - P(1)$$

First compute:

$$P(1) = \frac{e^{-0.2}(0.2)}{1!} = 0.2e^{-0.2}$$

$$= 0.1637$$

Thus:

$$P(X \geqslant 2) = 1 - 0.8187 - 0.1637 = 0.0176 \approx 0.0175$$

Q5. Poisson Distribution (λ = 3.5)

$$X \sim \text{Poisson}(3.5)$$

(a) At Least 2 Accidents

$$P(X \geqslant 2) = 1 - P(0) - P(1)$$

$$P(0) = e^{-3.5} = 0.0302$$

$$P(1) = 3.5e^{-3.5} = 0.1057$$

$$P(X \geqslant 2) = 1 - 0.0302 - 0.1057 = 0.8641$$

(b) At Most 1 Accident

$$P(X \leqslant 1) = P(0) + P(1)$$

$$= 0.0302 + 0.1057 = 0.1359$$

Q6. Poisson (λ = 3)

$$X \sim \text{Poisson}(3)$$

(a) $P(X \geqslant 3)$

Using complement:

$$P(X \geqslant 3) = 1 - P(0) - P(1) - P(2)$$

$$P(0) = e^{-3} = 0.0498$$

$$P(1) = 3e^{-3} = 0.1494$$

$$P(2) = \frac{3^2}{2!}e^{-3} = \frac{9}{2}(0.0498) = 0.2240$$

$$P(X \geqslant 3) = 1 - 0.4232 = 0.5768$$

(b) Conditional Probability

$$P(X \geqslant 3 \mid X \geqslant 1) = \frac{P(X \geqslant 3)}{P(X \geqslant 1)}$$

$$P(X \geqslant 1) = 1 - P(0) = 0.9502$$

$$= \frac{0.5768}{0.9502} \approx 0.607$$

Q7. Gaussian Distribution

Given PDF:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Comparing with:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

We identify:

$$m = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

Relations:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

---

1. $P(X \leqslant 0)$

$$= \Phi\left(\frac{3}{2}\right) = 1 - Q\left(\frac{3}{2}\right)$$

2. $P(X > 4)$

$$= Q(\frac{7}{2})$$

3. $P(|X + 3| < 2)$

$$-5 < X < -1$$

$$= \Phi(1) - \Phi(-1)$$

Using symmetry:

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

4. $P(|X - 2| > 1)$

$$= P(X < 1) + P(X > 3)$$

$$= 1 - Q(2) + Q(3)$$

Q8. Jury Decision (Binomial Model)

Let:

- Each juror correct with probability θ
- Conviction requires $\geqslant 8$ guilty votes.

If defendant is guilty:

$$\sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

If defendant is innocent:

$$\sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

If α = probability defendant is guilty:

$$\alpha \sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i} + (1-\alpha) \sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

Q9. Determining Constant c

Given:

$$p(i) = c\frac{\lambda^i}{i!}$$

Using normalization:

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using:

$$e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$$

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

(a)

$$P(X = 0) = e^{-\lambda}$$

(b)

$$P(X > 2) = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2}{2}e^{-\lambda}$$

Q10. Reliability of Systems

Number of functioning components:

$$X \sim \text{Binomial}(n, p)$$

System effective if $\geqslant$ half function.

(a) 5 vs 3 Components

5-component effective probability:

$$\binom{5}{3}p^3(1-p)^2 + \binom{5}{4}p^4(1-p) + p^5$$

3-component effective probability:

$$\binom{3}{2}p^2(1-p) + p^3$$

Inequality reduces to:

$$3(p-1)^2(2p-1) > 0$$

Thus:

$$p > \frac{1}{2}$$

(b) General Case

Difference:

$$P_{2k+1} - P_{2k-1} = \binom{2k-1}{k}p^k(1-p)^k(2p-1)$$

Since binomial coefficient and powers are positive:

$$> 0 \;\Leftrightarrow\; p > \frac{1}{2}$$

End of Lecture Scribe

This scribe follows exactly the derivations, definitions, assumptions, and reasoning provided in the lecture slides and solution document, structured step-by-step for exam revision.